

Bernoulli Distribution

Events that have one trial and two possible outcomes.

$$\text{Mean} = p$$

$$\text{PMF} = \begin{cases} q = 1-p & \text{if } k=0 \\ p & \text{if } k=1 \end{cases}$$

$$\text{CDF} = \begin{cases} 0 & \text{if } k < 0 \\ 1-p & \text{if } 0 \leq k < 1 \\ 1 & \text{if } k \geq 1 \end{cases}$$

$$\text{Median} = \begin{cases} 0 & \text{if } p < 1/2 \\ [0,1] & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$$

$$\text{Mode} = \begin{cases} 0 & \text{if } p < 1/2 \\ 0,1 & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$$

$$\text{Variance} = p(1-p) = pq$$

Without replacement
 After 1 white drawn:-
 9 white + 10 black left (19)
 Now Black removed
 10 white + 9 black
 $P(W_2|W_1) = \frac{9}{19} \approx 0.4737$
 $P(W_2|B_1) = \frac{10}{19} \approx 0.5263$
 Probability of success is not constant

8 balls are drawn from bag containing 10 white and 10 black balls. Predict whether the trials are Bernoulli trials if ball drawn is replaced and not replaced.

a) When ball is drawn with replacement:

$$\text{probability of success} \therefore p = \frac{10}{20} = \frac{1}{2}$$

b) When ball is drawn without replacement:

$$P(\text{Success}) \text{ for trial 1} = \frac{1}{2}$$

$$\text{for trial 2} = \frac{9}{19}$$

Not Bernoulli trial

With Replacement

$$P(W_1) = 10/20 = 0.5$$

$$P(W_1 \text{ and } W_2) = 0.5 \times 0.5 = 0.25$$

Binomial Distribution

2 possible outcomes: True or False, Success or Failure, Yes or No

Every trial is independent trial which means the outcome of one trial does not affect the outcome of another trial

$${}^nC_x = \frac{n!}{x!(n-x)!} \quad \# \text{ of experiments}$$

$P \Rightarrow$ Probability of success

$Q \Rightarrow$ Probability of failure $\Rightarrow 1 - p$

$$\text{Mean} \Rightarrow \mu = np$$

$$\text{Variance} \Rightarrow \sigma^2 = npq$$

$$\text{SD} \Rightarrow \sigma = \sqrt{npq}$$

Consider 3 Bernoulli trials for tossing a coin. $H \rightarrow$ Success $T \rightarrow$ Failure

3 ways in which we can have 2 success in 3 trials $\{SFF, FSF, FFS\}$

No. of successes for the above cases can take 4 values 0, 1, 2, 3

$a \rightarrow$ Prob of success $b \rightarrow$ Prob of failure

$$P(X=0) = P(FFF) = P(F) \times P(F) \times P(F) \\ \Rightarrow b^3$$

$$P(X=1) = P(SFF, FSF, FFS) = P(S) \times P(F) \times P(F) + P(F) \times P(S) \times P(F) + P(F) \times P(F) \times P(S) \\ = 3ab^2$$

$$P(X=2) = 3a^2b$$

$$P(X=3) = a^3$$

$$(a+b)^3 \Rightarrow a^3 + 3ab^2 + 3a^2b + b^3$$

Coin tossed 8 times

$\rightarrow$ Exactly 5 heads

$$P(a) = 1/2 \quad P(b) = 1/2$$

$$P(X=5) = {}^8C_5 a^5 b^{8-5} = \frac{8!}{3!5!} \times (1/2)^5 \times (1/2)^3$$

$$= \frac{8!}{3!5!} \times (1/2)^8 = 219/256$$